

Recording a route without satellites: speed from vibration, direction from turns, and their failure modes

Yu Yao-Hsing
euler.yu@gmail.com

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Abstract

A phone can record where it has been without satellite positioning, reading speed from how the vehicle shakes and direction from its motion sensors. We describe the method as shipped in a consumer iOS workout application and measure it against GPS recorded alongside, used only as a reference, over journeys by car, motorcycle and on foot. Driving fast and steadily with the phone resting in the vehicle, distance is within 3.8% over 15 km; in slow stop-start traffic the same system first read 20–60% long. Most of that gap was a caching policy rather than the signal: the store kept rare fast examples by evicting from its most crowded speed bin, which left it holding 5% of its examples below 5 km/h where the riding produced 24%. Replacing a random example instead, and calibrating by percentiles, left later rides roughly unbiased on average, though single rides still vary by tens of percent. Direction needs no satellite at all: the angle at which the phone sits is learned from the sideways acceleration of the vehicle's turns, and walking is steered by the axis of the steps. Compared route by route with GPS over 60 recordings and 451 km, the whole route is rotated by a median 13° and within 30° on 88% of recordings, against 31° and 48% for the phone's compass alone and 73% for earlier builds that learned the angle from GPS. With rotation and size removed all three draw the same shape: the gyroscope gets the shape right, and what separates a good route from a bad one is how far the whole of it is turned and how long it is. We report in full over twenty approaches that were implemented, measured and rejected — among them a sequence network that beat the store offline and was withdrawn after it failed in the field — and the platform semantics that cost more than any modelling choice: a vendor sensor fusion that removes the signal aircraft dead reckoning needs, a platform heading that goes singular in a pocket, and an acceleration reported with the opposite sign to the vehicle's.

1 Introduction

Satellite positioning fails where routes are most often wanted: tunnels, underground car parks, dense urban canyons, and aircraft cabins. The standard response is inertial dead reckoning — integrate acceleration twice — which is well known to diverge, since accelerometer bias is indistinguishable from sustained real acceleration and its double

integral grows without bound.

An alternative exists for wheeled vehicles. Road and engine vibration scale with speed, so a spectral signature of vertical acceleration can be mapped to speed directly, without integration and therefore without accumulation. CarSpeedNet [1] reports RMSE 1.8 m/s and MAE 0.72 m/s from 13.2 hours of accelerometer-only driving, at the same 4-second window length we arrive at independently. Its 0.5-hour test holdout is drawn from that same collection, so the reported figure describes journeys much like the ones it learned from.

This paper describes such a system built into a shipping application, and — more usefully — reports what happens outside the distribution, on a phone that is sometimes mounted, sometimes pocketed, sometimes in a hand, in a car, on a motorcycle, and once in an aircraft.

Contributions.

1. An out-of-distribution evaluation of vibration-derived speed across carry positions and vehicle types, on one device and one user, with per-regime error.
2. A cache-policy bias: an eviction rule that protects rare fast examples silently reshapes the distribution the estimator answers from, accounts for most of the low-speed over-read, and is fixed by changing the eviction, not by reweighting at lookup (§4.3).
3. Direction without GPS — a heading that cannot go singular, the carry angle from cornering, walking direction from the step axis — and a route-level comparison with GPS over 60 recordings that separates each route’s error into rotation, size and shape (§4.4, §4.5, Appendix A).
4. The measured cost of the specific-force ambiguity on a consumer platform: two approaches defeated, and a third we had blamed on it withdrawn.
5. Negative results in full, including a sequence network that beat the store offline and was withdrawn in the field, and two approaches that look correct in aggregate and fail per query.
6. Platform and systems defects that silently invalidated data — among them a test mode that used the GPS it was meant to exclude, and four meanings of “heading” — offered as evidence that platform semantics dominate modelling error in this setting.

2 Related work

Speed from vibration. The closest work is CarSpeedNet [1], a CNN taking three-axis smartphone accelerometer input, reporting RMSE 1.8 m/s and MAE 0.72 m/s over 13.2 hours of driving. Related approaches estimate speed from chassis vibration with

CNNs [2], infer position directly from vehicle vibration with an IMU [3], and use learned momentary speed to constrain strapdown inertial navigation [4]. Outside the inertial domain, vehicle speed has been estimated from tyre-road noise, where band energies at 1–4 kHz give 1.4% average error [5] and the underlying physics — tread-block passage at $f = NV/L$ — gives a direct speed dependence.

Our estimator is deliberately simpler than these: a k -NN regressor over log-spectral band energies, chosen because it can *decline* to answer when nothing resembling the query has been observed, which a trained network cannot. The contribution here is not the estimator but the evaluation regime.

Published accuracy is usually measured on familiar data. CarSpeedNet’s 0.5-hour holdout comes from the same 13.2-hour collection as its training data. This matters: we measure +1.0 to +1.7 km/h bias on a store built from matched journeys and +6.8 km/h on the same device once the store contains other carry positions and vehicles — a degradation invisible to a random split. The transportation-mode literature has confronted this more directly; the Sussex–Huawei Locomotion dataset [6] was assembled specifically to enable reproducible cross-user evaluation, with benchmark F1 near 0.93 for eight-class mode recognition [7]. No equivalent public dataset exists for *speed* regression across carry positions.

Inertial dead reckoning. That double integration of consumer MEMS output is unusable is well established: laboratory characterisation of smartphone IMUs reports position error exceeding 1 km within 100 s from pure inertial propagation [8], with error growth dominated by uncompensated bias and varying substantially between handsets. Learned approaches to inertial odometry [9] and the broader survey of deep inertial positioning [10] attack this by replacing integration with regression, as we do. Standard mitigations — zero-velocity updates [11] and map matching [12] — bound rather than remove the drift.

Heading and carry position. Pedestrian dead reckoning divides into step detection, heading estimation and position propagation [13], and heading is consistently the limiting term. Carry position is known to dominate: measured heading standard deviations of 6.22° , 6.82° and 14.68° for holding, calling and swinging respectively, with corresponding final position errors of 2.11, 2.34 and 4.5 m [14]. Our β term is the vehicular analogue of the carry-mode misalignment that literature addresses for pedestrians. An earlier draft of this paper reported that β cannot be recovered without an external reference. That was wrong: a vehicle’s turns reveal it (§4.4).

The specific-force ambiguity. An accelerometer measures specific force and cannot separate gravity from sustained linear acceleration. This is standard, and attitude-estimation work explicitly compensates for it, either by estimating disturbance acceleration adaptively [15] or by weighting the accelerometer correction by how far its magnitude departs from g [16]. What we add is the downstream consequence on a platform where

the fusion is not ours to modify: the vendor-supplied attitude and gravity-removed acceleration are the only interface, and their handling of sustained acceleration silently defeats two otherwise reasonable designs (§5.2).

3 Method

3.1 Signature

Vertical acceleration $x[n]$ is sampled at 50 Hz into a ring buffer of $N = 256$ samples (4.0 s), windowed and transformed:

$$w[n] = \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2\pi n}{N-1}\right) \quad (1)$$

$$X[k] = \sum_{n=0}^{N-1} (x[n] - \bar{x}) w[n] e^{-2\pi i k n / N} \quad (2)$$

Energy is accumulated into nine logarithmically spaced bands bounded at 0.4, 0.8, 1.6, 2.5, 4, 6, 9, 13, 18 and 24 Hz:

$$f_b = \ln\left(\sum_{k \in b} |X[k]|^2 + \epsilon\right), \quad b = 1 \dots 9 \quad (3)$$

Two time-domain terms complete an eleven-dimensional signature:

$$f_{10} = \ln \sigma(x), \quad f_{11} = \ln\left(\frac{1}{N-1} \sum_n |x[n] - x[n-1]|\right) \quad (4)$$

The window length was measured rather than assumed. At 4 s the held-out bias is +1.0 km/h at 2–10 km/h; doubling to 8 s raises it to +6.0, because the window then straddles changes in speed. Finer low-frequency banding changed nothing.

3.2 Speed

No functional form is imposed. The system stores $(\mathbf{f}, v)$ pairs whenever GPS supplies a speed, and answers by locality. With running feature means μ_i and variances σ_i^2 ,

$$d^2(\mathbf{f}, \mathbf{g}) = \sum_{i=1}^{11} \left(\frac{f_i - \mu_i}{\sigma_i} - \frac{g_i - \mu_i}{\sigma_i} \right)^2 \quad (5)$$

$$\hat{v} = \frac{\sum_{j=1}^K w_j v_j}{\sum_{j=1}^K w_j}, \quad w_j = \frac{1}{d_j^2 + \epsilon} \quad (6)$$

with $K = 12$, where π_j corrects for how over- or under-represented a neighbour’s speed is in the store relative to how often that speed was actually travelled:

$$\pi(v) = \text{clamp}\left(\frac{P_{\text{ridden}}(\text{bin}(v))}{P_{\text{store}}(\text{bin}(v))}, 0.05, 20\right), \quad w_j = \frac{\pi(v_j)}{d_j^2 + \epsilon} \quad (7)$$

P_{ridden} is a decaying histogram of every observation as it was learned, over 2 km/h bins, with a half-life of about 1400 observations; P_{store} is the same histogram of what the store currently holds. The two differ because of the eviction policy, not because of anything about the signal, and §4.3 shows that difference accounts for most of the low-speed over-read. Both are 1 on a fresh install, where the estimator reduces to the unweighted form above. Crucially the estimator is allowed to decline:

$$\min_j d_j^2 > 4 \implies \text{no answer} \quad (8)$$

Error scales directly with match distance — MAE 5.3 km/h below $d^2 = 1.03$, rising to 16.9 in the range 2.12–4.36 — so refusing a distant match removes the worst readings.

The store holds 4000 observations. Once it is full, an incoming observation replaces one chosen at random, with two exceptions: the fastest 2% of ground speeds are never chosen, so the store keeps evidence for speeds it rarely sees, and a ground observation never replaces an airborne one. §4.3 explains why this replaced an earlier rule, and why π remains only as a small correction while an old store turns over.

Averaging twelve neighbours pulls every answer toward the middle of their speeds. A monotone calibration undoes that: over held-out stored observations, the q -th percentile of what the estimator predicts is mapped onto the q -th percentile of what GPS measured, at 21 points, and every answer is passed through that map. Finally the *displayed* speed is smoothed over 2 s; distance is integrated from the unsmoothed estimate, because smoothing lags a real change and costs a few percent of distance.

3.3 Regime separation

A store holding several vehicles holds several mutually incompatible mappings. Measured directly: a ground-trained model reported 19 km/h at a 900 km/h cruise, because cruise is *quieter* than taxiing and the slowest stored signature was the closest match. After flight data entered the same store, a city drive reported a mean of 337 km/h against a true 30. Air and ground are therefore stored and searched as separate partitions.

3.4 Direction

Turn angle comes from the gyroscope, absolute direction from the magnetometer:

$$\theta_t = \theta_{t-1} + \Delta\psi_{\text{gyro}} + \kappa(\psi_{\text{datum}} + \beta - \theta_{t-1}), \quad \kappa = 0.25 \quad (9)$$

where β is the *carry offset*: the angle between where the phone points and where the body or vehicle travels. Earlier versions learned β from GPS course during the first 180 s of a workout and then froze it. The current one uses no satellite information after the first point, in four parts.

A datum that cannot go singular. ψ_{datum} is not the platform’s heading. Core Motion reports the bearing of a direction fixed in the phone, and which direction depends

on how the phone is held; when it points near vertical the reading is noise (§6). Instead, with the attitude $R = R_z(\text{yaw}) R_x(\text{pitch}) R_y(\text{roll})$ mapping the device frame to north–west–up, we take the bearing of whichever device axis lies most nearly horizontal — the one least aligned with gravity, switching only when another is flatter by 0.15 in $|\cos|$ — and carry the difference between the two axes across a switch so the reading stays continuous. Only offsets are ever learned against it, so its absolute zero does not matter.

The carry offset from cornering. In a turn a vehicle accelerates toward the inside of the turn. Writing the horizontal acceleration in the datum’s frame as a complex number $a = a_{\text{fwd}} + i a_{\text{right}}$, with v the estimated speed and ω the gyroscope turn rate (clockwise positive),

$$a_t \approx e^{i\beta} (i v_t \omega_t) \quad \implies \quad \hat{\beta} = \arg \sum_t \frac{a_t}{|a_t|} \overline{i v_t \omega_t} \quad (10)$$

over every riding second at $v \geq 4$ m/s. Normalising a_t keeps one pothole from outvoting a minute of corners. The acceleration must be the physical one, which is the *negative* of what Core Motion reports as user acceleration (§6). The first estimate is made after 30 riding seconds.

Walking from the steps. With the phone in a pocket the compass was 56–137° from the direction of walking on five of the seven walks graded, so walking is steered by the steps instead. Over 4 s, the principal axis of horizontal physical acceleration is the line of travel, and its third moment along that axis is positive toward the front, because push-off is sharper than heel-strike. Each window’s vote enters a circular median of $(\text{vote} - \psi_{\text{datum}})$ over the last 20 windows, which picks the end of the current window’s axis; the heading is pulled toward it at 0.5 per second. A window counts only if the phone’s posture held within 15° across it, and settling at a new angle (25°) starts the record afresh. A phone lying flat ($|g_z| > 0.7$, in the hand) takes the end nearer the datum instead, because the vote came out reversed on the one flat walk recorded.

Redrawing. Every drawn point keeps the heading and offset it was drawn with. When a ride’s $\hat{\beta}$ moves by 3°, every point of the current ride is turned by the difference; once a walk has eight counted windows, its opening windows are redrawn with the settled median. The saved route therefore carries the best estimate reached, while the live display carries the one available at the time. A ride begins at 2.5 m/s, and at that moment the heading is set to $\psi_{\text{datum}} + \hat{\beta}$ outright, because sitting down turns the phone about a horizontal axis the gyroscope’s vertical rate does not see.

3.5 Projection

$$\Delta s = \hat{v} \Delta t, \quad \phi_t = \phi_{t-1} + \frac{\Delta s \cos \theta_t}{R}, \quad \lambda_t = \lambda_{t-1} + \frac{\Delta s \sin \theta_t}{R \cos \phi_t} \quad (11)$$

Only the first point of a route originates from GPS.

4 Results

Throughout this section *real* means the speed or distance GPS recorded at the same instant as the estimate, and *estimated* means what the app produced from the motion sensors alone. Real distance is the integral of GPS speed over time, not the sum of fix-to-fix jumps: on one 15 km journey those two differ by 35% (14,445 m against 19,496 m), because position scatter adds length that was never travelled.

4.1 Distance over a whole journey

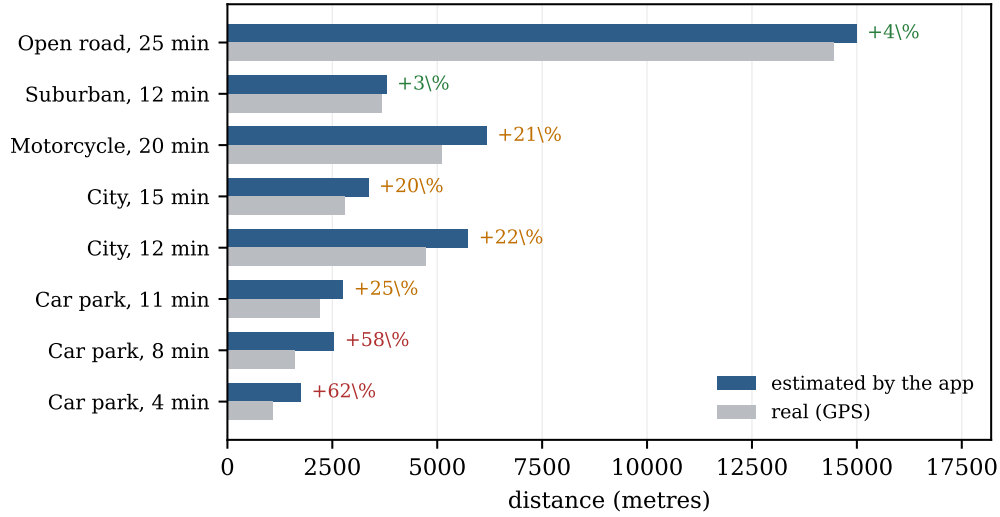


Figure 1: Estimated distance against real distance, eight journeys. The estimate is close when the vehicle is fast and steady, and reads long when it is slow.

Table 1: Every journey, in full. Average speed is over moving time only.

Journey	Estimated	Real	Error	Avg. speed
Open road, 25 min	14 994 m	14 445 m	+3.8%	43 km/h
Suburban, 12 min	3 793 m	3 678 m	+3.1%	28 km/h
Motorcycle, 20 min	6 181 m	5 118 m	+20.8%	35 km/h
City, 15 min	3 363 m	2 801 m	+20.1%	21 km/h
City, 12 min	5 741 m	4 720 m	+21.6%	29 km/h
Car park, 11 min	2 762 m	2 207 m	+25.2%	20 km/h
Car park, 8 min	2 548 m	1 616 m	+57.6%	20 km/h
Car park, 4 min	1 755 m	1 086 m	+61.6%	19 km/h
Motorcycle in a pocket	4 466 m	2 535 m	+76.2%	21 km/h

Table 1 lists every journey. The best is a 15 km open road drive at +3.8%; the worst road case is a four-minute car-park trip at +61.6%. The last row is the phone carried in a pocket rather than resting in the vehicle, which is a different failure and is discussed

in §8.

4.2 Why slow journeys read long

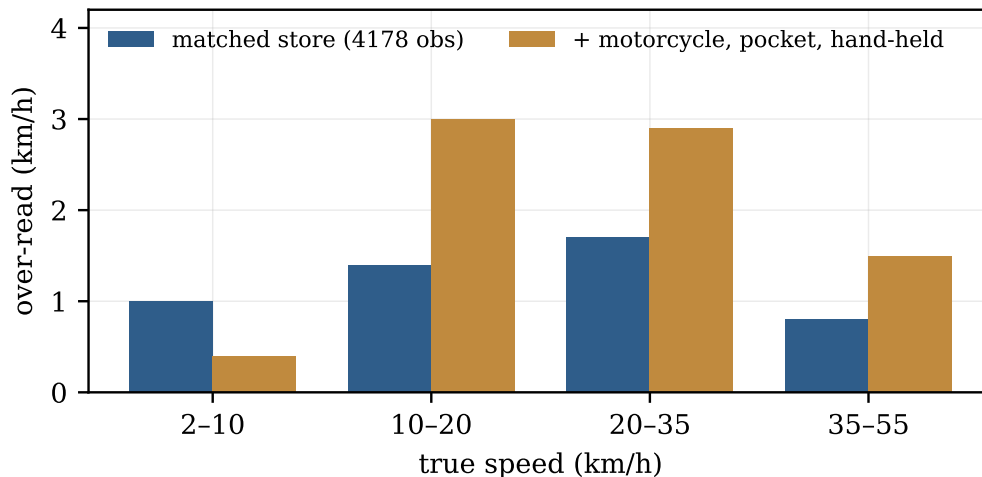


Figure 2: How much the estimate reads above the real speed, by real speed. Tested by hiding each stored example in turn and asking the model to recover it.

The ordering in Table 1 follows average speed, not any property of the route, and Figure 2 explains why. Tested against 4,178 stored examples from matched journeys — hiding each one in turn and asking the model to recover it — the estimate reads 1.0 to 1.7 km/h high at *every* speed.

A fixed error in km/h is not a fixed error in percent. Two km/h on a 50 km/h road is 2%; the same two km/h on a 5 km/h crawl is 40%. So the estimator that looks excellent on a motorway looks poor in traffic, without anything about it having changed.

This is also why no correction helps. Subtracting a constant 6 km/h — fitted on two journeys where the estimate happened to read exactly that much high — repaired those two and made three others worse, reducing total absolute error across all journeys by only 7%.

That account is correct as far as it goes, and it is not the whole story. It was measured by holding out stored examples, so it describes the error on the distribution the store holds. The riding has a different distribution, and §4.3 shows that the difference between the two — created entirely by the cache’s eviction rule — is the larger term.

4.3 The cache policy is the bias

The store holds 4000 observations. When it is full, an incoming observation displaces one drawn from the most crowded 2 km/h speed bin. The intent is to stop a rare 100 km/h

example being squeezed out by thousands of red lights, and it achieves that. What it also does is make the store stop resembling the riding.

Table 2: What was ridden against what the store keeps, over 44 instrumented recordings. The eviction rule is the only difference between the two rows.

	< 5 km/h	5–30 km/h	> 30 km/h
Observations as learned	24%	31%	45%
Observations the store keeps	5%	26%	69%

A query that lands between regimes is answered by whatever its twelve nearest neighbours happen to be, and after rebalancing most of them are fast. That predicts a specific signature of error: correct where the store is dense in both versions, wrong and high where the rebalancing has thinned the evidence. This is what the shipped application does — within a few percent above 50 km/h, roughly double at 15–30 km/h.

To test it rather than assert it, the estimator was reimplemented offline against the same feature extraction, the same K , the same decline threshold, and trained leave-one-ride-out on the other rides’ labelled windows. The only variable is how the training pool is assembled.

Table 3: Distance error, leave-one-ride-out. Rebuilding the pool under the shipped eviction rule reproduces the shipped error; the same estimator on an unbalanced pool does not. Weighting keeps the balanced store and corrects the imbalance at lookup.

Ride	Shipped app	Replay, pool built		
		evicted	natural	weighted
Motorcycle, 14 Sep	+32%	+7%	−5%	−4%
Motorcycle, 15 Sep	+53%	+23%	+3%	+8%
Motorcycle, 16 Sep a	+11%	−0%	−19%	−8%
Motorcycle, 16 Sep b	+39%	+38%	−2%	+10%
Motorcycle, 16 Sep c	+50%	+36%	+12%	+9%
Motorcycle, 16 Sep d	+30%	+15%	−9%	−8%
Mean error	36%	20%	8%	8%

Two things follow from Table 3. The estimator was never the problem: on a pool that was never rebalanced it is within 8% on the same rides, using the same eleven features that §5.4 had concluded carried no speed. And the fix does not require giving up the eviction rule — weighting each neighbour by $\pi(v)$ recovers the same 8% while keeping the rare fast examples an aircraft partition depends on. The residual gap between the replay’s 20% and the shipped 36% is the store’s older content, which the replay cannot reconstruct.

We record this as a modelling lesson of the same kind as the platform defects in §6: nothing about the eviction rule is wrong locally, it is sound cache management, and it was invisible for weeks because the store’s contents were never compared against the

distribution they were drawn from.

What shipped instead. The weighting of Table 3 did not survive contact with the field. The ridden histogram is a decayed count of the last few rides, while the store was built over months, and the ratio between the two is not the correction the store needs: the next build’s logs showed a median weight of 1.01 at 15–30 km/h where about 4 was needed, and replayed as shipped the weighting gave 22% against 22% without it. What converges is changing the eviction itself. Replayed from a store built under the crowded-bin rule through seven rides in order, that rule stayed at 5% slow observations throughout and the last four rides averaged 82% distance error; replacing a random observation instead reached 11% slow and 13% error over the same four rides, improving from the very next one. That is what ships (§3.2), with the weighting kept as a residual correction worth 13% $\rightarrow$ 11% in the same replay.

After it and the percentile calibration, whole-ride distance over eight later rides was roughly unbiased, -1% to -4% pooled. What remains is compression of the speed range: 5–15 km/h still reads about double and 50–80 km/h about a quarter low, and mean absolute speed error stays near 9.7 km/h whatever correction is applied — the floor set by the signature, not by the estimator. Figure 5(d) shows the same history in the recorded routes: drawn at a median 1.02 times their real size in August, 1.27 in the first half of September, and 0.98 in the last week.

4.4 Direction: the angle the phone sits at

The phone rarely points the way the vehicle travels. It sits at some angle in a cradle, a pocket or a cup holder, and that angle — β in §3.4 — has to be subtracted before the recorded heading means anything.

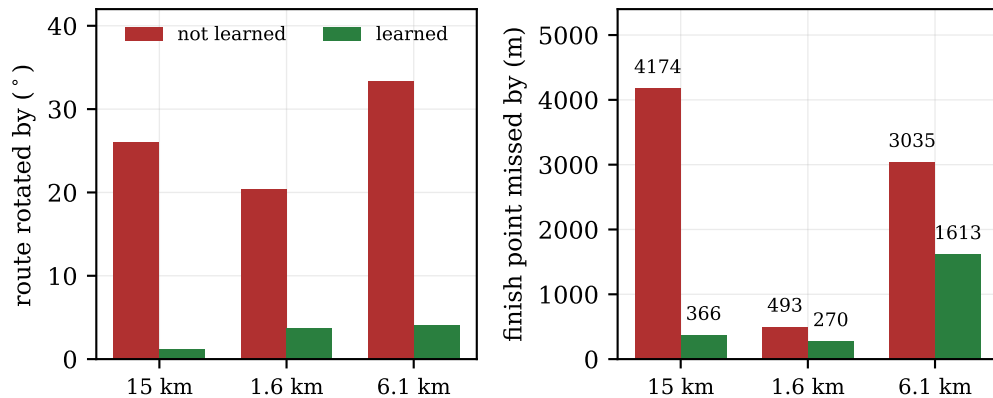


Figure 3: Why the angle matters, measured with the earlier GPS-learned version. The same journeys with the angle unlearned and learned. Left: how far the whole route is rotated. Right: how far the finish point lands from where it really was.

Figure 3 shows what that is worth. With the angle unlearned the route comes out the

right shape and the right length and simply points the wrong way: on the 15 km journey the recorded straight-line displacement matched the real one to within 8 m while being rotated 26° , which put the finish 4,174 m from where it actually was. Learning the angle moves the finish to 366 m, 2.4% of the distance travelled. Those figures come from the earlier version, which measured the angle from GPS in the first three minutes and then held it.

Without GPS. The current version learns the angle from turns and uses no satellite information after the first point (§3.4). To see what that costs, the direction logic was replayed through every recording that logged the phone’s attitude: 61 recordings from 15 August to 24 September, car and motorcycle, mounted and pocketed, 476 minutes of riding over 344 km. Direction is replayed; distance is left as each build recorded it, so the routes below carry whatever speed error was current at the time. GPS enters only as the grader: for every half-minute of riding, the bearing between the GPS fixes 15 s either side — at least 150 m and four times their accuracy apart, each better than 60 m — against the circular mean of the drawn heading over the same half-minute.

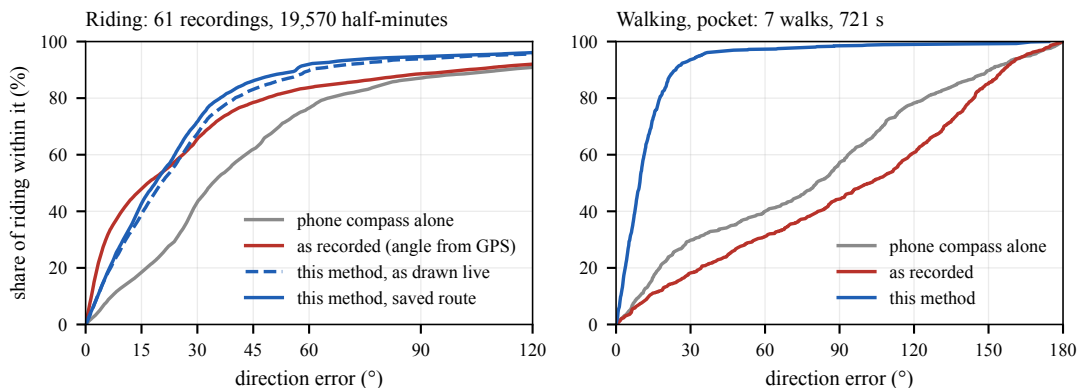


Figure 4: How far the drawn direction is from the real one. Left: 19,570 graded half-minutes of riding over 61 recordings. *As recorded* is what each build drew at the time, which for all but the last four recordings meant an angle learned from GPS. Right: 721 graded seconds of walking with the phone in a pocket, over seven walks.

Table 4 and Figure 4 summarise it. Without GPS the typical error is slightly worse than with the GPS-learned angle, 18.7° against 16.7° , but the failures are fewer: 72% of riding within 30° against 66%, and 3 recordings worse than 40° against 11. The GPS-learned angle is excellent when it is learned and wrong for the whole journey when it is not — frozen at a bad value from one poor fix, or never learned because the first three minutes were spent standing still. The compass alone is the floor, and in a pocket it is a poor one: within 30° for 30% of riding. Walking changes the most, because the recorded builds had no working walking direction at all.

Against the angle measured from GPS on the 27 rides where that measurement is possible — measured with the platform’s heading as the datum, before the change to the axis

Table 4: Direction error over every recording with attitude logged. Half-minutes of riding for the first four rows, seconds of walking for the last three. Pocket and flat split the 61 riding recordings by how the phone lay (35 and 26).

	Median	Within 30°			Recordings
	error	all	pocket	flat	over 40°
Riding, phone compass alone	34.1°	43%	30%	62%	24
Riding, as recorded (angle from GPS)	16.7°	66%	61%	72%	11
Riding, this method, live	20.3°	67%	68%	67%	7
Riding, this method, saved route	18.7°	72%	73%	69%	3
Walking, phone compass alone	80.8°	30%			
Walking, as recorded	102.3°	18%			
Walking, this method	9.6°	93%			

heading — the cornering estimate was 11.0° off (median) after 30 riding seconds and 9.4° at the end of the ride, within 30° on 26 of 27. Redrawing each ride with its latest estimate is what moves the saved route ahead of the live one. It is not free: on about ten recordings the later estimate was the worse one, and redrawing spread it back over the ride, by up to 24°.

4.5 Routes as a whole

A heading error per half-minute does not say what the finished route looks like. So each recording’s drawn route was compared with its GPS track as a whole, three ways: the phone’s compass alone, the route as recorded at the time, and this method — all three with the distance the phone recorded, so they differ only in direction. For every recording the drawn and GPS positions at the same instants were paired, and the rotation and scale about the shared start that best lay one on the other were found by least squares:

$$c = \frac{\sum_i \bar{z}_i w_i}{\sum_i |z_i|^2}, \quad \text{rotation} = -\arg c, \quad \text{size} = 1/|c|, \quad \text{shape} = \sqrt{\frac{\sum_i |c z_i - w_i|^2}{\sum_i |w_i|^2}} \quad (12)$$

with z_i and w_i the drawn and GPS positions as complex numbers east + i north. Rotation is the direction error of the route as a whole, size is the distance error, and shape is whatever is left: 0 is the same route turned and scaled. GPS fixes are those under two seconds old and better than 40 m, with impossible jumps removed; 60 recordings qualify, 451 km of GPS track.

Table 5 and Figure 5 give the result, and its most useful part is panel (c). Once rotation and size are taken out, the three versions of every route are equally close to the truth — a median shape difference of 0.10 to 0.12. The gyroscope draws the shape of a journey correctly whatever supplies the absolute direction; what separates a good route from a bad one is almost entirely how far the whole thing is turned, and how long it is.

On rotation the method without GPS is ahead of the builds that learned the angle from

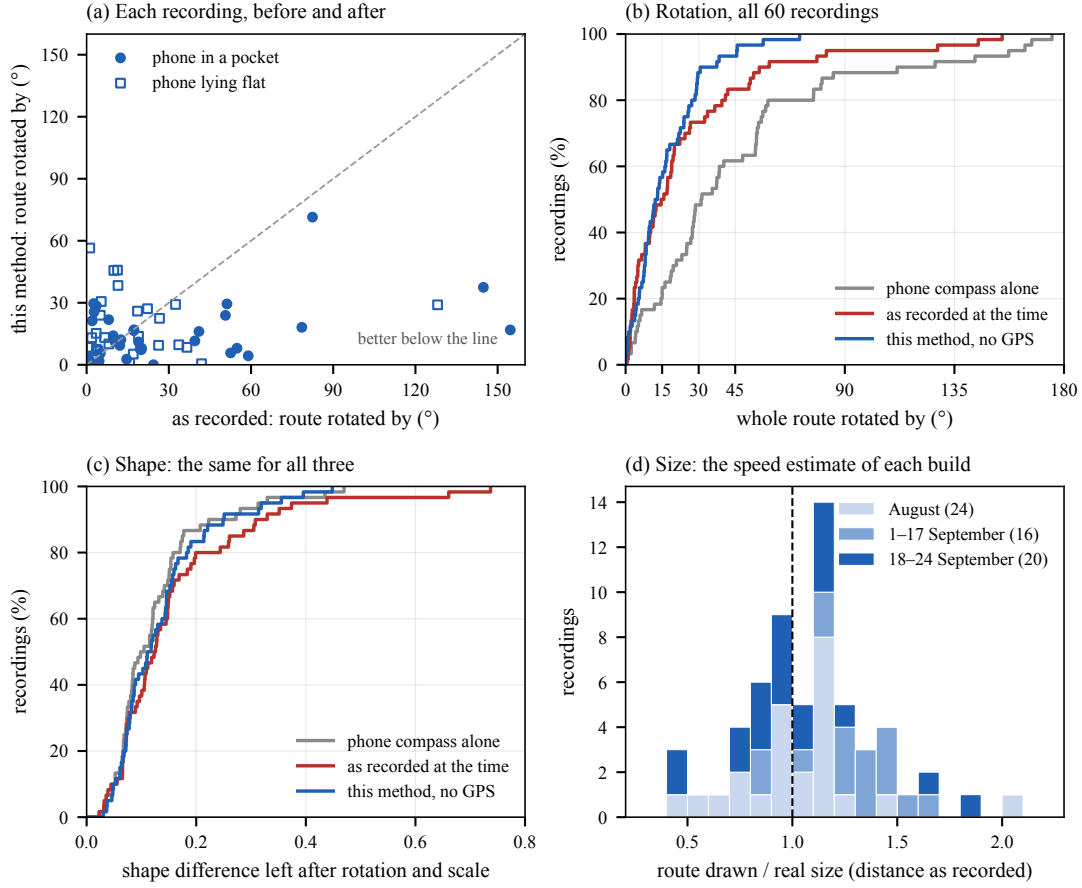


Figure 5: Every recording's route against its own GPS track. (a) How far each whole route is rotated, as recorded against this method; below the line is better. (b) The same as distributions, with the compass alone. (c) What is left once rotation and size are removed: the shape, which the gyroscope gets equally right in all three. (d) Route size, which is the speed estimate of the build that recorded it; direction does not change it.

Table 5: Route similarity over 60 recordings. Rotation and shape as defined above; drift is the distance from the GPS position at the same instant, divided by the distance travelled so far, median over each recording beyond its first 300 m, then median over recordings. Appendix A lists every recording.

	Whole-route rotation			Shape	Drift
	median	within 30°	over 45°	median	median
Phone compass alone	31°	48%	23	0.10	35%
As recorded at the time	15°	73%	10	0.12	24%
This method, no GPS	13°	88%	4	0.11	21%
<i>Phone in a pocket, 34 recordings</i>					
Phone compass alone	54°	26%	21	0.12	53%
As recorded at the time	17°	68%	9	0.13	29%
This method, no GPS	11°	94%	1	0.11	21%

GPS: whole routes within 30° on 88% of recordings against 73%, and four rotated past 45° against ten. With the phone in a pocket the gap is widest — 94% against 68%, one against nine — because that is where the GPS-learned angle failed outright when it failed, frozen for a whole ride at a value learned from one poor fix. The compass alone rotates a pocketed route by a median 54°. On the 26 recordings with the phone lying flat the two are level (81% within 30° each), and there the compass alone is already within 30° on 77%.

Size is not a direction property and is the same in all three versions; panel (d) shows it by period, and §4.3 explains the history. Both still matter for how far a point lands from where it was: a median rotation of 13° on its own moves a point by $2 \sin(6.5^\circ)$, about 23% of its distance from the start, and the median size error of 12% by 12%.

4.6 Underground car parks

Table 6: The two car-park journeys split by the ramp detector. A ramp is where the estimate fails; the road between the ramps is ordinary.

Journey	Where	Estimated	Real	Error
Car park, 4 min	on the ramp	267 m	67 m	+299%
Car park, 4 min	on the road	1 488 m	1 018 m	+46%
Car park, 8 min	on the ramp	618 m	229 m	+170%
Car park, 8 min	on the road	1 930 m	1 387 m	+39%

The two worst journeys in Table 1 are both car parks, and Table 6 locates the failure. A car crawling up a concrete spiral shakes as hard as one doing 40 km/h on a road, so the estimate on a ramp runs 170% to 299% long while the road between the ramps behaves ordinarily.

Detecting a ramp needs two signals together. The barometer alone flags every hill —

8 km of one 15 km journey, where the estimate is fine. Rotation alone flags every junction. A helical ramp is the only thing that climbs steadily *and* turns continuously the same way, so the app withholds distance when the barometer shows more than 0.08 m/s of sustained climb and the gyroscope more than 150° of same-direction turning within 30 s.

5 Negative results

We report these in full because each cost real effort and two share one cause.

5.1 Aircraft speed cannot be worked out at all

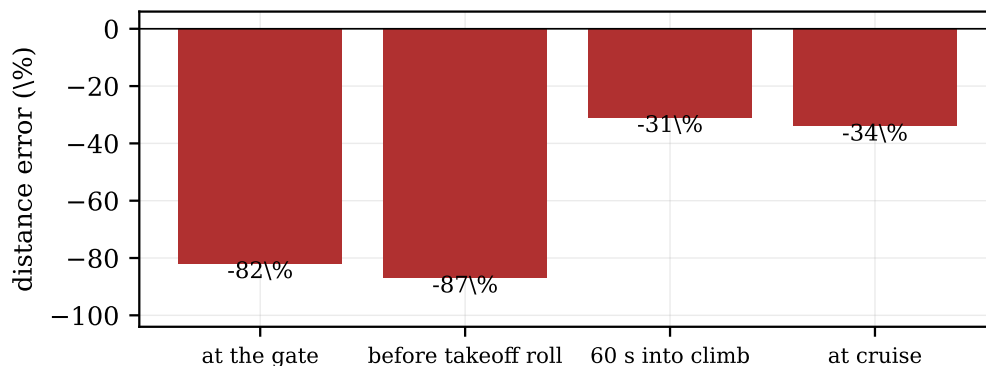


Figure 6: A real 22.8-minute flight replayed with GPS removed at four points, giving the distance the app would have recorded had signal been lost there.

Table 7: The same flight, in numbers. Signal is normally lost on the ground, which is the worst case: the last speed the app saw was taxi speed, and it holds it.

GPS lost	Speed held	Estimated	Error
At the gate	5.6 km/h	14.7 km	-82%
Just before takeoff	15.9 km/h	10.2 km	-87%
60 s into the climb	320.7 km/h	49.7 km	-31%
At cruise	307.6 km/h	46.5 km	-34%

A flight was recorded with GPS valid on all 1323 ticks, reaching 706 km/h. Replaying it with GPS removed (Figure 6, Table 7) shows the problem clearly: signal is normally lost on the ground, so the last speed the app has is taxi speed, and it holds it for the whole flight.

The correction that should rescue this — adding up the acceleration along the direction of travel — recovers about 60 km/h of a 679 km/h climb. After 400 seconds of held speed it had contributed -2 km/h. Direction, by contrast, is unaffected: 3.9° median across the same flight. The route points the right way and is the wrong length.

5.2 One cause, three failures

Core Motion’s attitude filter reinterprets sustained linear acceleration as a change in the gravity direction. The ambiguity itself is standard [15, 16]; what is reported here is its measured cost when the fusion is vendor-supplied and cannot be replaced. A takeoff roll is the most sustained acceleration a phone will ever experience, and is therefore precisely what the filter removes. This defeated two designs, and we had wrongly blamed it for a third:

1. *Aircraft speed integration*, above.
2. *Deriving β from acceleration* — withdrawn. A vehicle accelerates along its axis, so the world-frame acceleration bearing should give the travel direction; measured on three drives, circular concentration was $R = 0.15, 0.18, 0.19$, and we attributed that to the fusion. Most of it was ours. Core Motion’s user acceleration is the reaction the phone feels, the negative of the vehicle’s acceleration, so every bearing was reversed; and the frame was built by guessing between the attitude matrix and its transpose, which a phone lying flat cannot tell apart. Corrected, straight-line speed changes still give a poor answer — 44° median error — but the sideways acceleration of turns gives the carry angle to about 11° (§4.4).
3. *Inertial takeoff detection*. Integrating positive along-track acceleration and latching at 45 m/s in one unbroken episode should identify a takeoff and nothing else. Replayed: today’s car 52.5 m/s (latched), a second car 209.9 (latched), a motorcycle 328.5 (latched), a two-minute *walk* 119.7 (latched), and the actual flight 15.4 — never latched. Backwards on every case. The episode rarely breaks in a vibrating vehicle because road noise holds the signal above the floor, so the integral accumulates the positive half of the noise indefinitely.

5.3 Regime identification does not transfer

Whole-session fingerprints — the mean normalised signature of a workout — separate regimes cleanly. Across eight sessions, same-vehicle-and-carry pairs sit 0.54–1.70 apart, same-vehicle-different-carry 1.81–2.87, and different-vehicle 4.30–6.97.

Restricting the neighbour search by that fingerprint nevertheless made prediction *worse* ($+2.2 \rightarrow +3.4$ km/h). The ranking is unreliable at the individual level: for one car journey the nearest stored session was a car (1.38), but the second was a *hand-held* session (1.54) and the third a *motorcycle in a pocket* (1.74), both ahead of the same car driven the same morning (2.29).

5.4 On a motorcycle in a pocket the signature carries no speed at all

Every failure above is a matter of degree: the estimate is too high, or too low, or right about the wrong regime. This one is different in kind. Over a 19.1-minute motorcycle ride with the phone in a trouser pocket — 1146 ticks at a clean 1 Hz, Velocity Mode on for 1143 of them, three GPS fixes in the drawn route — the reported speed and the

real speed are statistically unrelated: $R = +0.13$. The vibration signature and the real speed are related at $R = -0.02$, which is to say not at all.

Table 8 shows what that looks like from the outside. The estimate reports roughly 50 km/h whatever the motorcycle is doing, including when it is stationary at a light.

Table 8: Motorcycle, phone in a pocket. Mean estimated speed within each band of real speed. The estimate is flat: it does not fall when the motorcycle stops and does not rise when it accelerates.

Real speed	Ticks	Mean estimated
0–2 km/h (stopped)	47	36.2 km/h
2–10 km/h	25	54.4 km/h
10–25 km/h	3	57.8 km/h
25–45 km/h	14	55.5 km/h
45–70 km/h	54	49.8 km/h

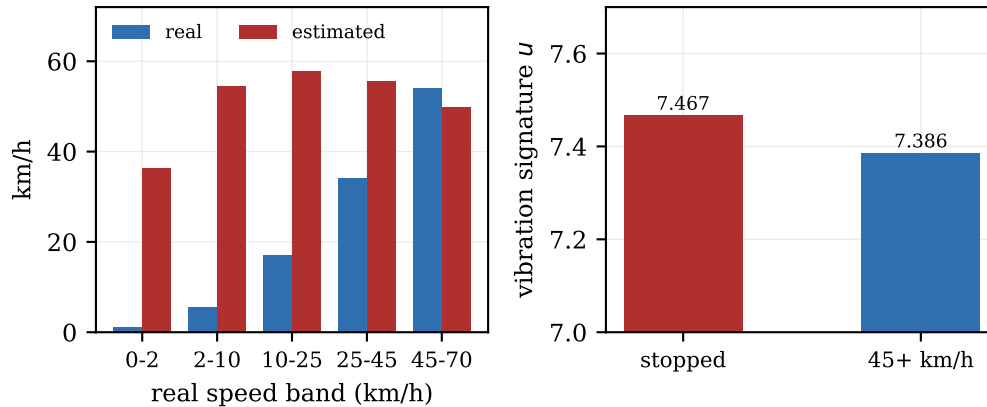


Figure 7: Left: estimated against real speed by band — the estimate is flat where the truth is not. Right: the vibration signature is indistinguishable between a standstill and 45 km/h, and marginally *higher* at rest.

The cause is visible in the signature itself. Its median value is 7.467 while the motorcycle is stopped and 7.386 while it is doing over 45 km/h — indistinguishable, and marginally higher at rest. A car carries the phone on a rigid surface and delivers road and tyre noise that scales with speed. A motorcycle delivers engine vibration, which tracks engine speed rather than road speed and is undiminished at a standstill in gear, through a layer of clothing that damps what little road input survives. There is no speed information in the input, so no estimator built on this feature can recover one; the 47 stationary ticks accumulated 468 m of distance that did not happen.

What makes this the most serious result in the paper is not the error but the confidence. Fingerprint distance sat at 3.51 for the entire ride, outside the same-vehicle-different-carry range of 1.81–2.87 measured in §5.2 and approaching the different-vehicle range.

The extrapolation flag — the mechanism whose job is to notice exactly this — was *false on all 1146 ticks*. The system reported 53.8 km/h at a red light and never signalled that it was answering about a regime it had never seen.

5.5 Correction: the pocket signature does carry speed

§5.4 is the strongest claim in this paper and it is partly wrong. The quantity measured there at $R = -0.02$ against real speed is a scalar from the superseded hand-built estimator — a scaled dominant vibration frequency, which the diagnostics file still records beside the learned model’s output — and not the eleven-dimensional signature §3.1 defines. The two were conflated. Re-measured on four motorcycle rides with the phone in a trouser pocket under open sky, the signal is present and strong:

Table 9: Motorcycle, phone in a trouser pocket, 16 September. RMS of vertical acceleration against real speed, in 4-second windows. The relationship the earlier scalar could not see.

Real speed (km/h)	0–5	5–15	15–30	30–45	45–60	60–90
Median RMS (m/s ²)	0.11	0.39	0.63	0.95	1.11	1.27

Over 331 windows on one ride, RMS ranks with real speed at Spearman +0.82, ahead of the shipped application’s own reported speed (+0.73); spectral centroid ranks at -0.64 , so the information is in amplitude rather than in where the energy sits. RMS is already one of the eleven features (f_{10} , §3.1). The flat ~ 50 km/h reading that §5.4 reports is therefore not an absence of signal in the input but the cache bias of §4.3 overwhelming it: the same features, searched against an unbalanced pool, place the same pocketed rides within 8–10%.

What survives from §5.4 is the part about confidence, which the correction makes worse rather than better. The system reported 53.8 km/h at a red light, the extrapolation flag was false on all 1146 ticks, and the diagnosis that followed — *this carry has no signal* — was itself wrong and stood for three weeks, because the log recorded a feature that was no longer the one being used. An instrument that reports a superseded quantity beside a current one, with no indication of which is which, produces confident wrong conclusions in the analyst as readily as in the application.

5.6 Confidence signals that do not work

Neighbour disagreement is the natural confidence measure for a k -NN regressor. It fails here in the wrong direction: out-of-distribution motorcycle queries had a *tighter* neighbour spread than in-distribution car queries (5.2 vs 12.4 km/h), because they land consistently in one wrong region. Session-level fingerprint distance correlated with speed bias at $r = -0.01$ over six journeys.

5.7 Discriminating a hijacked pedometer

On a motorcycle, engine vibration is counted as footsteps. Apple’s activity classifier reported **automotive** on 636 ticks and **walking** on 271 of one continuous ride, and each flicker allowed the pedometer to take over: it drove 27% of the ride, reporting a mean 3.3 km/h while GPS recorded 49.6.

Three per-tick discriminators were measured against real walking and rejected: vibration energy (79% caught at 21% false), gait-band spectral concentration (distributions overlap), and both combined with rotation rate (69% at 7%). What worked was history rather than instantaneous features: across two real walks the classifier *never* reported automotive in 190 stepping ticks, while on motorcycles the median gap since it last did was 84 s. Requiring two minutes of silence before a pedometer-driven revocation suppressed 61% of false hand-overs at zero measured cost.

5.8 Four more that were measured and rejected

Integrating between stops. Each detected stop is a known zero, so acceleration along the travel direction can be integrated between two of them with the drift pinned out at both ends. Measured over the segments of one ride: –65% of the real distance using the application’s heading, –50% using the raw motion heading, and –75% even when the GPS course was supplied as a perfect direction. Residual drift reaches 10–21 m/s per segment for a pocketed phone.

Time since the last stop. If the over-read were a pull-away artefact it would concentrate after a stop. It does not: on two rides the excess distance splits +64%, +54%, +48% and +81% across 10–30, 30–60, 60–120 and 120 s+ since the last stop.

Zeroing an idling engine. Shake does not separate a stationary motorcycle with its engine running from one crawling: accelerometer magnitude has median 0.22 m/s² stopped against 0.65 at 4–12 km/h, heavily overlapped. The best threshold tested removes 1142 m of phantom distance across 70 recordings while zeroing 312 crawling and 200 slow-riding ticks — about 2000 m that was really travelled.

Reweight the amplitude features. Since f_{10} and f_{11} carry the speed information (§5.5), weighting them above the nine band energies in the distance metric should help. It changes nothing measurable (8% mean distance error at $\times 1$, $\times 2$ and $\times 3$). The neighbour search was never the problem; the pool was.

5.9 A neural network: better offline, withdrawn in the field

The k -NN store is simple on purpose (§2), so the obvious question is whether a trained network does better. Seven designs were trained on the same eleven features and scored on held-out rides, each given the same percentile calibration and the same gates, with mean absolute speed error in km/h:

1. A per-window CNN in place of the store: worse, 11.1 against 9.4 on the ground and 88 against 34 in the air.

2. The average of the two: worse than the better one alone.
3. A CNN given the store’s calibration and gates: closer, never ahead.
4. Spectrum input, augmentation and ground pretraining: the best per-window network, and still behind.
5. A residual network on top of the store, initialised as an exact copy of it: 11.0 against 10.8, no gain.
6. *Distillation*: a student trained on the store’s own answers matches it exactly, 10.9 against 10.9, and needs no GPS labels at all — the store’s function compresses into 40,000 parameters.
7. A *sequence* model, a GRU over 20–40 s of features: 8.8 against 10.6 for the store, and 9.2 for the store with its 2 s display smoothing. Doubling context, layers, data or seeds bought 0.1 km/h. Time, not capacity, is what the store lacks.

The sequence model was then shipped for eleven builds, running beside the store on every tick with a developer setting choosing which one drew the route and both answers logged. On two car drives it did what the offline numbers promised — 2.8 against 3.2 km/h on 570 paired ticks — and something they had not measured: the store had no answer on 32% of one drive, because a car is far from a store built on a motorcycle, while the network answered 95% of the time. Its cost was not a reason to hesitate either: 1.9 ms and 0.31 J for a whole ride against 1.1 ms and 0.10 J for the store. A recurrent layer cannot run on the Neural Engine; a convolutional version that can was 4.7 km/h against the GRU’s 3.6 on the same split, to save 0.2 J.

It was withdrawn after a retrain. The retrained network read a pocketed motorcycle ride 34% low where its predecessor had read 13% low, plausibly because its calibration had been fitted on a training set by then dominated by 8,493 windows of car-mounted motorway driving. The next ride its owner judged a complete failure, and every build since has run the store alone. The cause of that ride is not known; its log was never sent for analysis.

A larger lesson came from the same work, on a different network. A small CNN on the raw 50 Hz window beat the store’s per-tick error on ten of eleven held-out rides, 7.9 against 14.1 km/h, and left distance error by speed band exactly where it was, because the compression comes from averaging under uncertainty, not from the features. Out of distribution it failed unsafely: on a flight it had never seen it averaged 223 km/h of error and recorded 5% of the distance, where the store declines to answer. A network that is better on familiar data and confidently wrong on unfamiliar data is the wrong trade for a system whose purpose is the unfamiliar case.

5.10 Six ways of finding direction that did not work

Each was built, replayed against GPS on the logged rides, and dropped. Numbers are median direction error unless stated.

The axis the motorcycle leans about. A leaning bike rolls about its own direction of travel, so the principal axis of horizontal rotation should give that direction up to a half-turn. It gives 25° even with the half-turn forgiven.

The thigh as a pointer. Walking, gravity in the phone’s frame runs down the thigh; seated, the thigh points forward, so the same device direction should point along the road. It does not: seated, that direction stayed $54\text{--}86^\circ$ from horizontal, and the estimate was 73° off.

A lean-proof forward vector. Leaning is a rotation about the forward direction, so solving for that direction in the phone’s own frame — least squares on $\mathbf{a}_{\text{dev}} = -v\omega [\mathbf{u}]_{\times} \mathbf{f}$ — should be immune to lean. It scored 19.9° against 15.9° for the planar method over 29 rides, better on six and worse on nine. Lean is not what limits the estimate.

Gating the compass by the gyroscope. Following the gyroscope and letting the compass move the baseline only while the two agree within 30° improved seven rides by more than 10° and worsened five; riding within 30° fell from 69% to 65%. It was treating a symptom whose cause was the heading definition (§6).

Switching headings by conditioning. Using the platform’s heading while its reference direction is well off vertical and the axis heading otherwise made the ride it was designed for worse, 72° to 78° : every hand-over copies the noisier source into the offset.

The thigh’s swing, for walking forward or back. A thigh swings forward faster than it returns, so rotation about the axis across the direction of travel should be skewed toward the front. It was wrong in 56–62% of two- to six-second windows, because the largest horizontal rotation of a pocketed phone while walking is about the direction of travel, not across it.

6 Systems failures

The first three defects below were not algorithmic, and each silently invalidated data for weeks.

A stale reading is not a missing reading. `CLLocationManager` suspends heading updates outside the foreground, and the cached property is never cleared. On an 81-minute drive the heading datum held *one value for 4790 of 4832 seconds* while the device physically rotated 5264° . A fallback to Core Motion’s heading had been added for the missing case and was never reached, because it was written as `clHeading ?? motionHeading` and a frozen value is not `nil`.

Background permission gating. `allowsBackgroundLocationUpdates` was enabled only under *Always* authorisation. Under *When In Use* the process was suspended mid-journey: a single 371-second hole in a 21-minute drive, during which the 50 Hz stream delivered 49 samples where 18,550 were due, accounting for the entire 20% distance shortfall.

Instrumentation that cannot express the failure. A 17-minute workout the user believed was recording without GPS had the mode active for 34 seconds; 63 GPS fixes entered the route. The guarantee held — the code path returns early — but nothing in the log stated the mode’s state, and it took an hour to establish from a side effect. Two columns now record it directly.

A repeated reading is not a new one. The diagnostics file carries the latest GPS speed, accuracy and position on every tick, holding the last fix until another arrives — in Velocity Mode, up to 45 s. One ride held a 0 km/h reading for 33 consecutive ticks while the rider pulled away, which scored as the estimator reporting 60 km/h at a standstill and put that ride at +47% where fresh fixes alone give +36%. The column was not wrong; it was undated. Every figure in §4.3 is computed against fixes under two seconds old, using an age column added afterwards.

Four meanings of “heading”. The direction work was held back for weeks by platform semantics rather than by the estimator. Core Motion’s heading is the bearing of a direction fixed in the phone, chosen by how the phone is held — fitted as $(-0.05, 0.87, 0.49)$ in the device frame on some rides and $(-0.02, 0.46, 0.89)$ on others — and when that direction points near vertical the reading is noise. On one ride the phone sat head-down at 45° , the direction was within 20° of vertical, and over each minute the heading scattered 49° around what the gyroscope measured; the route was drawn 72° off. The attitude itself was fine: the most nearly flat device axis scattered 1° over the same minutes. **CLHeading** follows yet another direction, $25\text{--}100^\circ$ from Core Motion’s on a tilted phone, and the application handed over between the two whenever it entered or left the foreground; seven rides whose carry angle had looked like zero turned out to have been measured against **CLHeading**. Core Motion’s user acceleration is the reaction the phone feels, the negative of the vehicle’s acceleration: on pocketed rides it pointed $139\text{--}180^\circ$ away from the acceleration GPS measured. And a 30 s running mean of the vertical rotation rate, subtracted as gyroscope bias, is a high-pass filter that hands back every turn over the following minute — Core Motion’s rotation rate is already bias-corrected. The compass correction hides that last one, and removing it changed nothing measurable, so it remains.

A gate that cannot recover. The estimator refuses to answer when the workout’s fingerprint sits further than 3.0 from every learned session (§3.3). That fingerprint is the mean of the workout’s own GPS-labelled observations, and it only grows while GPS supplies a usable speed. On one ride GPS did so twenty times, during stops and a pull-away, and then effectively stopped: the fingerprint froze at 6.12 — a different-vehicle value, for the same motorcycle that had scored 0.29–1.88 that morning — and the gate refused every tick for 9.5 minutes, recording 20 m of a 4.3 km ride. Across 21 sessions the fingerprint exceeds 3 only while a session holds about twenty observations (3.35, 4.47, 4.78, each settling to 0.7–1.1) and never passes 2.57 after sixty. A refusal now requires sixty. The defect is not the threshold but that a quantity computed from an eight-sample minimum was allowed to be authoritative and could not be revised once the evidence feeding it stopped.

A test mode that used GPS. Velocity Mode can be forced on with GPS still available, to measure what it would do without. Twice that test quietly used GPS for direction. First, the carry angle was corrected continuously from GPS course — on 716 of 720 ticks of one drive — and later, after that was removed, it was learned from GPS during a three-minute warm-up. Every heading figure from those builds was flattered, and nothing in the output said so. The warm-up was removed in favour of the cornering estimate (§3.4), and the log now records the learned-without-GPS angle in its own column.

A warm-up on the wall clock. That warm-up was three minutes from pressing Start, not three minutes of movement, so a rider who warmed the engine and put a helmet on spent it standing still. The split across the forced rides was total: with no moving seconds inside the window the angle came out -147° , -33° and $+130^\circ$; with 46–131 moving seconds, -1° , -3° and $+2^\circ$.

A walking angle applied to a ride. The angle is learned separately walking and riding, because a pocketed phone sits differently in each. A correction that rotates the opening stretch of a route once the angle is known did not check which kind of angle it had: on five of 30 rides the first to settle was learned walking away from the bike, and it rotated the whole ride by $+146^\circ$, $+161^\circ$, $+127^\circ$, $+73^\circ$ and -55° .

A placeholder averaged as a measurement. Estimated points carry a nominal 250 m accuracy meaning “dead reckoned”. The workout’s average accuracy included them, so 6% of estimated points turned a real 12 m average into 26 m and a quality score of 16/100 — which we took, for a while, as the receiver having degraded. The best-instrumented drive, 2127 real fixes, has a median of 14.2 m and 97% within 20 m. The receiver was fine.

One point, two kilometres. A single dead-reckoned point once landed 2052 m from the GPS fix beside it in time, adding 5.4 km to a 49 km recording, while the other 131 sat a median 79 m from the truth. Nothing bounded a step by what the vehicle could have covered since the last one.

Both of the last two were fixed, and both fixes were reverted along with the network (§5.9) when the application was returned to its earlier build wholesale; neither is in the current version. We report them because a revert of that kind is itself a systems hazard: it restored the estimator that was wanted and silently removed two unrelated corrections with it.

7 Limitations

- **Hand carry destroys the signal; pocket carry degrades it.** At matched speeds a hand-held phone reads 0.6–0.8 lower in the superseded summary feature, which stops varying with speed altogether. A pocket is different and was misdiagnosed for three weeks (§5.5): the signature there does track speed, and a pocketed ride places within 8–10% once the store’s composition is corrected for.

- **Stationary vehicles with running engines.** A motorcycle at a red light reads 5.2 km/h, accumulating 690 m over 339 stationary ticks. No tested signal separates an idling engine from a slow crawl.
- **Car-park ramps.** A vehicle crawling a concrete spiral shakes as hard as one at 40 km/h on a road; measured error is +169% to +297%. Detection requires two signals jointly — barometric climb above 0.08 m/s *and* signed net rotation above 150° over 30 s — since the barometer alone flags every hill and rotation alone every junction.
- **Straight ramps are not detected, and it matters less than expected.** The turn requirement excludes any ramp that does not spiral. The first straight one recorded — 12.6 m of climb over 1.15 km at 45–74 km/h, under 45° of net turning — showed why the rule should not simply be relaxed: the estimate there was already right (1135 m against 1153 m by GPS), so withholding it would have discarded a kilometre of correct distance. A turn-free climb test tightened to 6 m of rise at $R^2 \geq 0.9$ still fires on 2169 ticks across 64 recordings, almost all ordinary hills, and on 131 ticks of a flight’s cabin pressure. Only a *slow* straight ramp needs catching, and slow is the one thing the estimate cannot vouch for without GPS.
- **Unseen vehicles.** The model answers only for conditions it has observed.
- **Speed held through a stop.** When the phone is taken out of a pocket the vibration is the hand’s, so the last speed is held rather than measured. Stopping and reaching for the phone therefore keeps the route moving: on one ride 48 km/h was held for ten seconds after the motorcycle had stopped, about 130 m that was never travelled.
- **The ramp detector at speed.** A rising, turning road can look like a car-park ramp. On one ride the detector withheld nine seconds at 48 km/h on an ordinary road, about 115 m.
- **Walking pace.** Step cadence times a learned stride read 4–7 km/h on one walk where GPS measured 3.4–4.7.
- **A phone that moves in the pocket.** The carry angle is learned per ride and assumed to hold. Where the phone slowly rolled in a pocket during a ride the error grew to 30°, and redrawing a ride with a later, worse estimate spread it back over the whole ride. A walk that begins while the phone is still being put away is drawn from the compass until the phone settles.

8 Threats to validity

One device, one user, three vehicles. The 23 journeys of §4.1 and the 60 recordings of §4.5 were all made on a single iPhone by one person, in one city, using one car, one

motorcycle and walking. Every quantity here should be read as characterising that configuration. The transportation-mode literature moved to multi-user public datasets [6] precisely because single-subject results do not transfer, and we have no evidence ours do.

GPS as ground truth. Errors are measured against GPS recorded simultaneously. Where GPS is degraded — underground, and in the urban canyon immediately after — the reference is itself uncertain. For walking it can be worse than uncertain: on one 4.5-minute walk the Doppler speed averaged 1.06 m/s while position chords gave 1.6–2.2 m/s depending on the sampling interval, at 14–20 m accuracy. The accelerometer settles it without GPS — autocorrelation gives a 1.05 s stride period, so the application’s 1.55 m/s implies a 0.85 m stride and the Doppler figure implies 0.55 m, which is not the gait that was recorded. Walking figures in earlier drafts that rested on GPS alone should be treated as unproven. We use the integral of GPS speed rather than the sum of fix-to-fix displacements, because the latter accumulates position scatter: on one 15 km drive the two differ by 35% (14,445 m against 19,496 m), and the fix-summed figure is the wrong one. Segments with no valid GPS speed are excluded and reported separately, but a systematic bias in GPS speed at low speed would propagate into our low-speed figures undetected.

GPS as the direction reference. Direction is graded on half-minute chords between fixes that are often 30–60 m accurate, which puts a floor under every figure: every variant of the direction logic we replayed scored between 18° and 21° median error over all riding, however different its failures. Recordings worse than 40°, where the reference cannot be the explanation, are the more telling measure. Walking truth needs a straight half-minute of good fixes and exists for only seven walks.

No baseline comparison. We do not evaluate a published method on this data, so we cannot say whether a CNN trained on the same journeys would degrade the same way outside its distribution. The comparison we do make — matched store versus mixed store on identical held-out data — isolates store composition, not architecture.

Thresholds fitted on few journeys. The ramp detector (0.08 m/s climb with 150° signed rotation over 30 s) was selected by sweeping against seven journeys and choosing the lowest total absolute error. That is a small sample, and one journey regressed from −7.1% to −31.7% under the chosen setting because its ramp over-read had been cancelling an under-read elsewhere. We report it rather than tuning it away.

Data availability. The logs are per-tick CSV plus 50 Hz raw traces, and the results here are reproducible from them. They are not currently public. Releasing them would make this a labelled multi-vehicle, multi-carry vibration corpus, which to our knowledge does not otherwise exist; we consider that the most useful next step.

9 Conclusion

Vibration-derived speed is a sound basis for GPS-free route recording within its regime: fast, steady travel with the phone mechanically coupled to the vehicle, where we measure 3.8% distance error over 15 km. Outside that regime the failures are not gradual — a hand on the phone removes the signal, and an aircraft’s defining acceleration is erased by the platform’s own sensor fusion. Direction, by contrast, no longer needs a satellite reference at all. Learned from the vehicle’s own turns and from the steps when walking, it puts whole routes within 30° of their real orientation on 88% of 60 recordings, ahead of the builds that learned the angle from GPS, and 93% of walking within 30° against 30% for the compass alone. The shape of a route was never the problem; its rotation was, and its size still is.

We would emphasise four lessons over any result. First, the dominant error sources were not modelling choices but platform semantics: a suspended process, a stale cached heading, a heading that goes singular when the phone sits head-down, an acceleration with the opposite sign to the vehicle’s, an attitude filter making a defensible choice that destroys the signal of interest. Second, a component can be locally correct and globally biasing: the cache’s eviction rule is sound cache management, and it silently reshaped the distribution the estimator answers from (§4.3); where a system learns continuously and stores what it learns, the storage policy is part of the model. Third, a model that is better on familiar data and confidently wrong on unfamiliar data is the wrong trade when the unfamiliar case is the purpose (§5.9). Fourth, a test is only a test if it cannot quietly use the thing it is testing without: twice the GPS-free mode used GPS for direction, and every figure from those builds was flattered.

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A Every recording

Every recording with the phone’s attitude logged and enough GPS to compare, in date order. *Size* is the drawn route’s size relative to the GPS track, set by the speed estimate of the build that recorded it. *Rotation* is how far the whole route is turned relative to the GPS track, clockwise positive, as recorded at the time and with this method. *Shape* is what remains once rotation and size are removed, for this method. *Drift* is the distance from the GPS position at the same instant as a share of the distance travelled so far, median over the recording. Definitions are in §4.5. *Flat* means the phone lay flat, as in a car mount; *pocket* that it stood more upright, almost always in a trouser pocket on a motorcycle.

Table 10: Recordings 1–30, 15 August to 14 September.

#	Recording	Phone	GPS		Rotation		Shape	Drift	
			km	Size	recorded	this		recorded	this
1	15 Aug 21:32	flat	10.9	1.12	-128°	$+29^\circ$	0.07	65%	16%
2	15 Aug 22:06	flat	1.6	1.13	$+4^\circ$	$+7^\circ$	0.08	6%	7%
3	16 Aug 12:32	flat	9.0	1.00	$+2^\circ$	-8°	0.14	9%	11%
4	16 Aug 13:42	flat	4.4	1.12	-2°	-4°	0.05	4%	8%
5	17 Aug 20:21	flat	3.2	0.94	$+11^\circ$	$+46^\circ$	0.06	14%	45%
6	17 Aug 21:05	flat	3.1	0.92	$+5^\circ$	-31°	0.09	9%	41%
7	19 Aug 08:46	pocket	13.0	0.45	-41°	-16°	0.22	31%	30%
8	19 Aug 10:52	pocket	6.6	0.57	$+8^\circ$	$+22^\circ$	0.31	62%	64%
9	19 Aug 18:15	flat	2.1	0.94	-22°	-27°	0.25	18%	19%
10	19 Aug 19:42	flat	1.6	2.06	$+11^\circ$	$+38^\circ$	0.32	40%	48%
11	21 Aug 20:55	flat	2.7	0.77	$+7^\circ$	$+13^\circ$	0.18	9%	11%
12	22 Aug 07:28	pocket	9.9	0.81	-4°	-29°	0.07	14%	33%
13	22 Aug 10:57	flat	9.5	0.77	$+34^\circ$	$+10^\circ$	0.14	29%	15%
14	22 Aug 20:42	flat	2.5	0.95	$+37^\circ$	$+8^\circ$	0.13	21%	6%
15	23 Aug 15:47	flat	10.4	0.66	-19°	-14°	0.25	29%	36%
16	23 Aug 18:38	flat	16.3	1.02	-27°	-22°	0.05	24%	21%
17	25 Aug 20:54	flat	1.8	1.48	$+8^\circ$	$+10^\circ$	0.19	31%	25%
18	25 Aug 22:48	flat	1.2	1.28	$+19^\circ$	$+26^\circ$	0.21	35%	39%
19	28 Aug 17:00	pocket	5.1	1.18	$+79^\circ$	-18°	0.10	72%	21%
20	28 Aug 17:57	pocket	4.8	1.01	-24°	-0°	0.19	21%	8%
21	29 Aug 10:40	flat	4.7	1.15	$+10^\circ$	$+46^\circ$	0.16	26%	60%
22	29 Aug 12:45	flat	4.9	1.18	-4°	$+8^\circ$	0.06	13%	17%
23	29 Aug 19:56	flat	2.9	1.20	-2°	-13°	0.04	14%	18%
24	31 Aug 08:06	pocket	12.3	1.17	-3°	$+0^\circ$	0.15	12%	12%
25	10 Sep 16:03	flat	8.5	1.21	$+16^\circ$	$+1^\circ$	0.17	18%	14%
26	11 Sep 07:37	pocket	13.8	1.68	$+17^\circ$	-17°	0.08	41%	42%
27	11 Sep 15:19	pocket	14.7	1.36	$+17^\circ$	-16°	0.12	22%	19%
28	11 Sep 21:50	flat	6.5	1.05	-1°	$+56^\circ$	0.03	4%	75%
29	14 Sep 08:02	pocket	11.9	0.88	-19°	$+11^\circ$	0.12	26%	12%
30	14 Sep 16:01	pocket	7.3	1.34	-40°	$+12^\circ$	0.09	43%	21%

Table 11: Recordings 31–60, 15 to 24 September.

#	Recording	Phone	GPS		Rotation		Shape	Drift	
			km	Size	recorded	this		recorded	this
31	15 Sep 08:06	pocket	11.9	1.55	+3°	+26°	0.07	41%	55%
32	15 Sep 13:05	pocket	9.5	0.84	−51°	−29°	0.45	58%	33%
33	16 Sep 08:56	pocket	12.0	1.45	−59°	−4°	0.05	66%	26%
34	16 Sep 11:42	flat	8.0	1.32	−5°	−24°	0.07	24%	38%
35	16 Sep 15:21	pocket	2.0	1.41	+2°	−1°	0.12	30%	26%
36	16 Sep 15:26	pocket	7.3	1.21	+3°	+30°	0.07	11%	47%
37	17 Sep 08:03	pocket	9.3	1.23	−10°	−11°	0.16	22%	21%
38	17 Sep 12:30	pocket	8.8	1.12	−53°	+6°	0.11	64%	13%
39	17 Sep 13:33	pocket	5.7	1.47	−5°	+6°	0.11	18%	21%
40	17 Sep 14:59	flat	6.5	1.12	−17°	−5°	0.15	17%	10%
41	18 Sep 09:00	pocket	12.0	1.26	−55°	−8°	0.08	50%	14%
42	18 Sep 12:33	pocket	4.1	1.61	−1°	+4°	0.07	43%	43%
43	18 Sep 15:17	pocket	16.6	0.99	−5°	−2°	0.09	15%	4%
44	18 Sep 19:12	pocket	8.0	1.84	−155°	+17°	0.16	41%	15%
45	19 Sep 17:47	pocket	4.0	0.80	+20°	−8°	0.08	26%	15%
46	19 Sep 21:00	pocket	3.2	0.42	+10°	−14°	0.36	32%	45%
47	20 Sep 11:34	flat	2.8	1.20	+4°	−15°	0.08	15%	34%
48	20 Sep 11:53	flat	2.8	0.97	+32°	+29°	0.04	44%	40%
49	20 Sep 13:17	flat	15.6	0.77	+26°	−9°	0.06	22%	15%
50	20 Sep 16:19	pocket	4.3	0.93	+2°	−21°	0.09	12%	32%
51	21 Sep 16:05	flat	5.1	0.49	+42°	−1°	0.21	51%	32%
52	22 Sep 08:06	pocket	11.7	0.92	−145°	−37°	0.15	98%	46%
53	22 Sep 13:00	pocket	6.0	1.18	+20°	−7°	0.15	15%	8%
54	22 Sep 15:38	pocket	13.9	0.85	+19°	+8°	0.13	11%	9%
55	23 Sep 08:41	pocket	11.5	0.87	−51°	−24°	0.11	36%	19%
56	23 Sep 11:50	pocket	5.7	1.14	−13°	−12°	0.14	12%	12%
57	23 Sep 12:57	pocket	1.7	1.07	+82°	+71°	0.40	42%	49%
58	23 Sep 15:14	pocket	13.1	0.85	−15°	−3°	0.15	28%	12%
59	24 Sep 07:58	pocket	10.4	1.14	−12°	−9°	0.05	13%	12%
60	24 Sep 16:07	pocket	10.3	1.00	+1°	−1°	0.09	3%	6%